

ML project lifecycle

Module 1 - Lesson 2

Overview

Machine learning projects follow a lifecycle: define the problem, gather and prepare data, train and evaluate, then deploy and monitor. Skipping steps causes silent failures.

Key Concepts

- Start with a well-defined business or product problem, not a model.
- Choose a metric that matches the decision the model supports.
- Prepare data: clean, label, split into train/validation/test sets.
- Train baseline models first, then iterate with better features and algorithms.
- Evaluate honestly on held-out data, not the data used to train.
- Deploy, monitor drift, and retrain as real-world data changes.

Example

```
Lifecycle checklist:  
1. Problem definition and success metric  
2. Data collection and labeling  
3. EDA, cleaning, and feature engineering  
4. Train/validate/test split  
5. Baseline -> iterate -> select  
6. Evaluate on test set  
7. Deploy, monitor, retrain
```

Summary

A disciplined lifecycle turns an idea into a working model. Define the metric, split data honestly, iterate, and monitor after deployment so the model stays correct.

Practice Checklist

- Write a one-page project brief: problem, metric, data source.
- Split a small dataset and confirm no leakage between sets.
- Sketch the monitoring plan for a deployed model.